

Experiment – 4

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(A)

You are given a table `tbl_happiness` that contains the ranking of happiness for different countries. Your task is to retrieve all the records from the table while ensuring that "India" and "Sri Lanka" appear at the top, followed by the rest of the countries in their original ranking order.

Expected Output: (i) part

Rankings	Country
126	India
128	Sri Lanka
1	Finland
2	Denmark
3	Iceland
4	Israel
5	Netherlands
6	Sweden
7	Norway

Solution:

```
CREATE TABLE tbl_happiness
(
    sno INT PRIMARY KEY,
    rankings INT,
    country VARCHAR(50)
);
```

```
INSERT INTO tbl_happiness
VALUES
(1,1,'Finland'),
(2,2,'Denmark'),
(3,3,'Iceland'),
(4,4,'Israel'),
(5,5,'Netherlands'),
```

```
(6,6,'Sweden'),
(7,7,'Norway'),
(8,126,'India'),
(9,128,'Sri Lanka');
```

```
SELECT RANKINGS, COUNTRY
FROM TBL_HAPPINESS
ORDER BY
    CASE
        WHEN COUNTRY = 'India' THEN 0
        WHEN COUNTRY = 'Sri Lanka' THEN 1
        ELSE 2
    END,
RANKINGS;
```

(B)

Make all the rankings in descending order except India and Sri Lanka.

(ii) Part

Rankings	Country
126	India
128	Sri Lanka
7	Norway
6	Sweden
5	Netherlands
4	Israel
3	Iceland
2	Denmark
1	Finland

Solution:

```
CREATE TABLE tbl_happiness
(
    sno INT PRIMARY KEY,
    rankings INT,
    country VARCHAR(50)
);
```

```
INSERT INTO tbl_happiness
VALUES
(1,1,'Finland'),
(2,2,'Denmark'),
(3,3,'Iceland'),
(4,4,'Israel'),
(5,5,'Netherlands'),
(6,6,'Sweden'),
```

```
(7,7,'Norway'),
(8,126,'India'),
(9,128,'Sri Lanka');
```

```
SELECT RANKINGS, COUNTRY
FROM TBL_HAPPINESS
ORDER BY
    CASE
        WHEN COUNTRY = 'India' THEN 0
        WHEN COUNTRY = 'Sri Lanka' THEN 1
        ELSE 2
    END,
RANKINGS DESC;
```

(C)

A company maintains employee information in a single table where each employee may report to another employee acting as their manager. Prepare a report that displays the details of every employee who supervises at least one other employee. The report should include the manager's employee ID, name, total number of direct reports, and the average age of those reporting employees. The average age should be rounded to the nearest whole number and the final result should be sorted by employee ID.

employee_id	name	reports_to	age
9	Hercy	NULL	43
6	Alice	9	41
4	Bob	9	36
2	Winston	NULL	37

Output

employee_id	name	reports_count	average_age
9	Hercy	2	39

Solution:

```
CREATE TABLE Employees
(
    employee_id INT PRIMARY KEY,
    name VARCHAR(50),
    reports_to INT,
    age INT
);

INSERT INTO Employees (employee_id, name, reports_to, age)
VALUES
(9, 'Hercy', NULL, 43),
```

```
(6, 'Alice', 9, 41),  
(4, 'Bob', 9, 36),  
(2, 'Winston', NULL, 37);
```

```
SELECT  
    E1.EMPLOYEE_ID,  
    E1.NAME,  
    COUNT(E2.EMPLOYEE_ID) AS REPORTS_COUNT,  
    ROUND(AVG(E2.AGE),0) AS AVG_AGE  
FROM EMPLOYEES E1  
JOIN EMPLOYEES E2  
ON E1.EMPLOYEE_ID = E2.REPORTS_TO  
GROUP BY E1.EMPLOYEE_ID, E1.NAME  
ORDER BY E1.EMPLOYEE_ID;
```